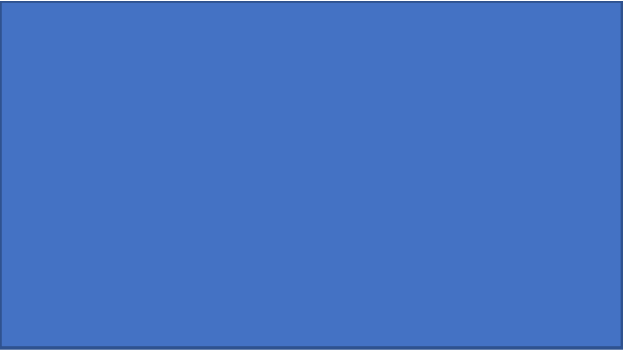




Methods for NDN based data transfer in multi-path networking environments

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Problem Description

Typical multi-path networks

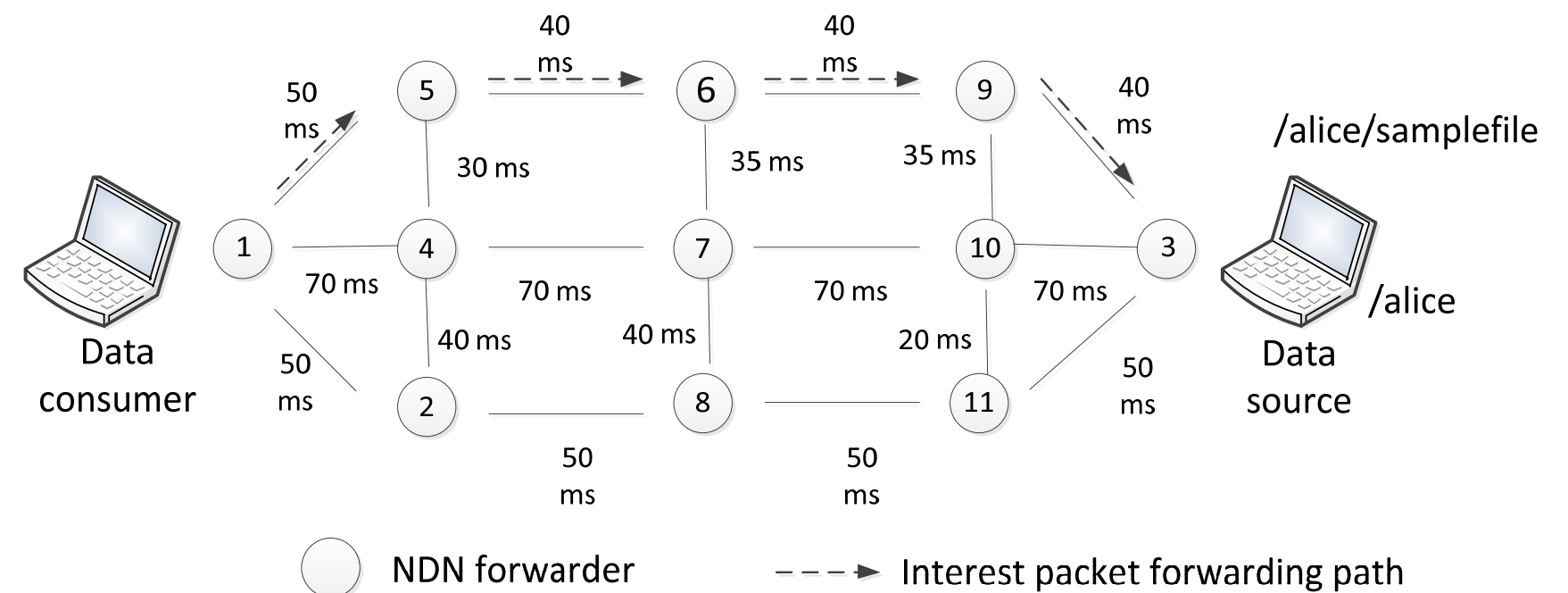
- Satellite communication (SATCOM) networks (LEO, MEO, GEO)
- Combination of multiple networks (SATCOM networks, LTE networks, Terrestrial networks)

Objectives

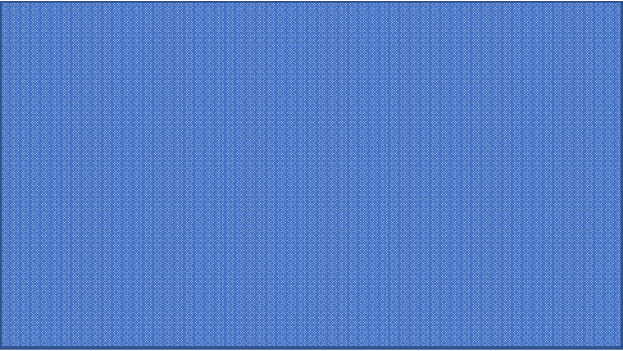
- Effectively utilize network throughputs in a multi-path NDN network (throughput increase)
- Improve data transfer robustness

Challenges

1. Multi-path delay differences
2. Unknown available throughputs in different network paths
3. Coupling of multi-paths between data consumer and data producer



Data transfer in a multi-path NDN network

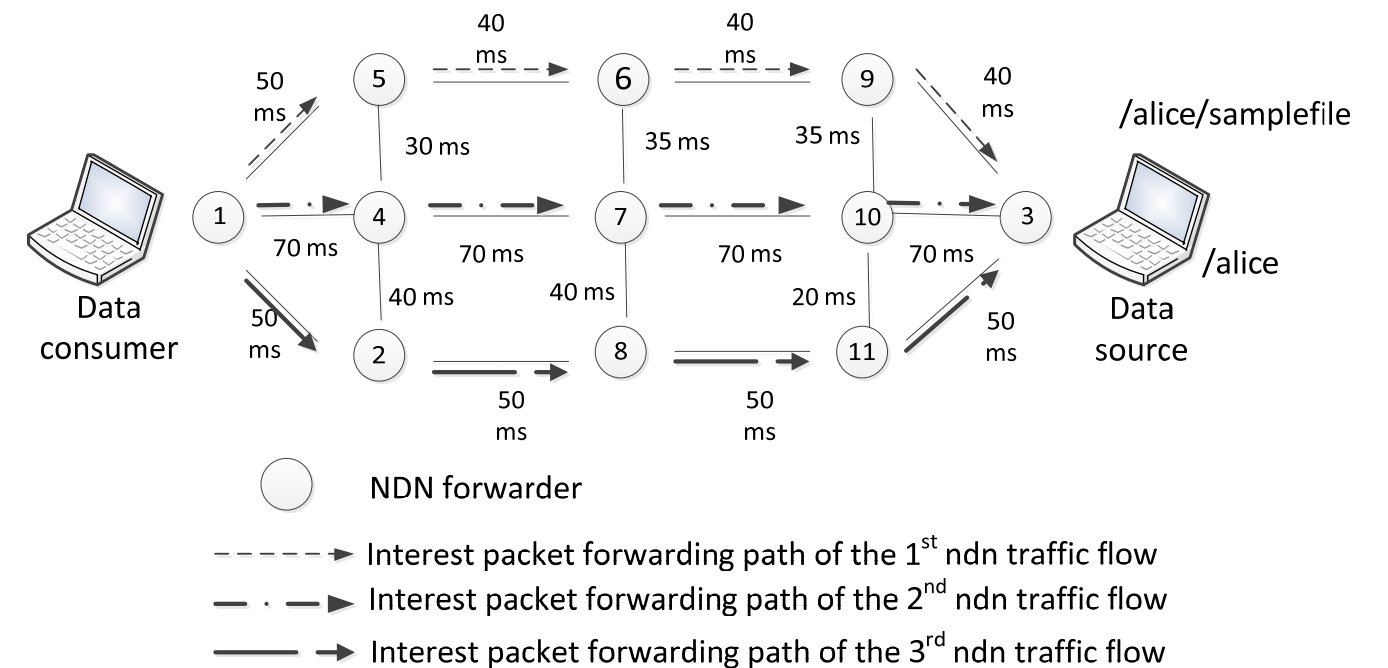


Solution 1

- **NDN forwarders support multipath routing**
 - Forwarders determine how to distribute NDN interest packets to multiple faces
 - Forwarders try to match incoming rate of interest packets with data packet returning rate from faces
- **NDN data consumer:**
 - Single congestion control is used at data consumer side for retrieving NDN data packets
- **Challenges**
 - Data consumer side congestion control handles round trip delay variations
 - Forwarding decisions are more complex
 - Effective throughput of the multi-path network depends on interest packet forwarding decisions

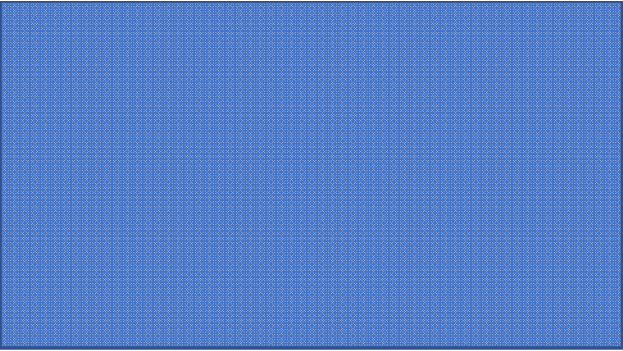
Solution 2

- **NDN data consumer:**
 - Add a flow ID field in NDN interest packets.
 - NDN data consumer assigns flow ID to interest packets to create a virtual NDN traffic flow for data transfer.
 - Each virtual NDN data flow (with the same flow ID) has its own congestion control.
- **NDN forwarders:**
 - NDN interest packets with the same file name, and different flow IDs may be forwarded differently by NDN forwarders.
 - NDN interest packets with the same file name and the same flow id will be forwarded to the same Face by NDN forwarders.
- The virtual NDN traffic flows are created sequentially by the data consumer.
- For each new NDN traffic flow, NDN forwarders perform a route discover process to identify additional throughputs in the multipath NDN network for data transfer.



Flow ID converts a multi-path network environment into a multiple single path network environment

- Delay variation in multiple network paths are addressed
- Forwarding decisions at the forwarders are simple.
- Available network throughput from each single path is not affected by forwarder forwarding decisions.
- Data consumer side congestion controls can effectively utilize throughputs in each network path



Thanks!